

B.TECH 1ST YEAR | 1ST SEMESTER

ENGINEERING MATHEMATICS

Unit II

Eigenvalues, Eigenvectors & Orthogonal Transformation

Complete Theory

40+ Solved Problems

50 Exercises

Cayley-Hamilton Theorem

Quadratic Forms

Exam Ready

Every step explained from basics — No shortcuts — Zero confusion

Designed for weak students and toppers alike

★ Step-by-step ★ Background explanations ★ Exam-oriented ★

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SECTION 1: INTRODUCTION TO EIGENVALUES AND EIGENVECTORS

1.1 What is a Matrix Transformation?

Before understanding eigenvalues, you must understand what a matrix does. A matrix $\mathbf{A}$ is a mathematical object that *transforms* vectors. When you multiply a matrix $\mathbf{A}$ with a vector $\mathbf{x}$, you get a new vector $\mathbf{Ax}$. This new vector is usually in a completely different direction from $\mathbf{x}$.

For example, if $\mathbf{x} = [1, 0]$ and $\mathbf{A}$ rotates every vector by 45 degrees, then $\mathbf{Ax}$ will point in a different direction. However, there are *special vectors* for every matrix that do **not** change direction when multiplied by the matrix — they only get scaled (stretched or compressed). These special vectors are called **eigenvectors**.

1.2 Definition of Eigenvector

Definition: A non-zero vector $\mathbf{x}$ is called an *eigenvector* of matrix $\mathbf{A}$ if multiplying $\mathbf{A}$ by $\mathbf{x}$ gives back the same vector $\mathbf{x}$ multiplied by a scalar λ . That is:

$$\mathbf{A} \cdot \mathbf{x} = \lambda \cdot \mathbf{x}$$

where λ (lambda) is a scalar called the **eigenvalue** corresponding to the eigenvector $\mathbf{x}$.

1.3 What Does This Mean Physically?

Think of it this way:

- Normally, when a matrix acts on a vector, it changes both the magnitude and the direction.
- But for eigenvectors, the matrix **only changes the magnitude** (by a factor of λ), **not the direction**.
- If $\lambda > 1$: the vector gets stretched in its own direction.
- If $0 < \lambda < 1$: the vector gets compressed.

- If $\lambda < 0$: the vector gets flipped and stretched/compressed.
- If $\lambda = 0$: the vector gets mapped to the zero vector (the matrix is singular).

1.4 Simple Numeric Example to Build Intuition

Consider the matrix:

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Take vector $x = [1, 0]$:

$$A \cdot [1, 0] = [3 \times 1 + 0 \times 0, 0 \times 1 + 2 \times 0] = [3, 0] = 3 \times [1, 0]$$

So, $x = [1, 0]$ is an eigenvector with eigenvalue $\lambda = 3$. The vector only got scaled by 3 — its direction didn't change!

Now take vector $x = [0, 1]$:

$$A \cdot [0, 1] = [3 \times 0 + 0 \times 1, 0 \times 0 + 2 \times 1] = [0, 2] = 2 \times [0, 1]$$

So, $x = [0, 1]$ is an eigenvector with eigenvalue $\lambda = 2$.

1.5 Why Are Eigenvalues Important?

Eigenvalues and eigenvectors have enormous practical importance:

- **Structural Engineering:** Finding natural frequencies of vibration of structures
- **Electrical Engineering:** Analyzing resonance in circuits
- **Computer Science:** Google's PageRank algorithm uses eigenvalues
- **Data Science:** Principal Component Analysis (PCA) uses eigenvectors
- **Quantum Mechanics:** Energy levels of atoms are eigenvalues
- **Differential Equations:** Solving systems of ODEs

Key Takeaway: Eigenvalues tell you *by how much* a matrix scales a special vector.

Eigenvectors tell you *which direction* is that special direction. Together, they completely describe the most important geometric behavior of a matrix.

SECTION 2: THE CHARACTERISTIC EQUATION

2.1 Deriving the Characteristic Equation from Scratch

We start from the definition: $A\mathbf{x} = \lambda\mathbf{x}$

We want to find λ and $\mathbf{x}$. Let us rearrange:

$$A\mathbf{x} = \lambda\mathbf{x} \Rightarrow A\mathbf{x} - \lambda\mathbf{x} = 0 \Rightarrow A\mathbf{x} - \lambda I\mathbf{x} = 0 \Rightarrow (A - \lambda I)\mathbf{x} = 0$$

Here, I is the identity matrix (we must write λI , not just λ , because you cannot subtract a scalar from a matrix directly — you must first convert it into a matrix form).

2.2 Why Must the Determinant Be Zero?

We have the equation $(A - \lambda I)\mathbf{x} = 0$. We want a **non-zero** solution for $\mathbf{x}$ (eigenvector must be non-zero by definition).

From linear algebra, the equation $M\mathbf{x} = 0$ has a non-trivial (non-zero) solution only when the matrix M is *singular*, i.e., its determinant is zero.

FUNDAMENTAL CONDITION

For $(A - \lambda I)\mathbf{x} = 0$ to have a non-zero solution $\mathbf{x}$, we need:

$$\det(A - \lambda I) = 0$$

This equation is called the **Characteristic Equation** of matrix A .

2.3 What is the Identity Matrix I ?

Identity Matrix: For a 2×2 matrix: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. For a 3×3 matrix: $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

It has 1s on the diagonal and 0s everywhere else. Multiplying any vector by I gives the same vector back.

2.4 Forming $A - \lambda I$ Step-by-Step (2x2 Example)

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$.

So:

$$A - \lambda I = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}$$

2.5 Computing $\det(A - \lambda I)$ for 2x2

$$\det(A - \lambda I) = (a-\lambda)(d-\lambda) - bc = 0$$

Expanding:

$$\begin{aligned} ad - a\lambda - d\lambda + \lambda^2 - bc &= 0 \\ \lambda^2 - (a+d)\lambda + (ad-bc) &= 0 \\ \lambda^2 - (\text{trace of } A)\lambda + (\det \text{ of } A) &= 0 \end{aligned}$$

Memory Tip: For a 2×2 matrix, the characteristic equation is always:

$$\lambda^2 - (\text{sum of diagonal elements})\lambda + (\text{determinant of } A) = 0$$

The sum of diagonal elements is called the **trace** of A.

2.6 Characteristic Polynomial

The expression $\det(A - \lambda I) = 0$ gives a polynomial in λ . This polynomial is called the **characteristic polynomial** of A. Solving this polynomial gives us all the eigenvalues.

- For an $n \times n$ matrix, the characteristic polynomial has degree **n**
- A 2×2 matrix gives a degree-2 polynomial (quadratic) $\rightarrow$ 2 eigenvalues
- A 3×3 matrix gives a degree-3 polynomial (cubic) $\rightarrow$ 3 eigenvalues

SECTION 3: FINDING EIGENVALUES

3.1 Step-by-Step Process to Find Eigenvalues

To find eigenvalues of a matrix A , follow these steps:

1. Write the matrix A clearly.
2. Form the matrix $A - \lambda I$ by replacing each diagonal element a_{ii} with $(a_{ii} - \lambda)$.
3. Compute the determinant of $(A - \lambda I)$.
4. Set this determinant equal to zero: $\det(A - \lambda I) = 0$.
5. Solve the resulting polynomial equation for λ .
6. Each value of λ you find is an eigenvalue.

3.2 Determinant of a 3x3 Matrix — Full Expansion

For a 3×3 matrix $M = [[m_{11}, m_{12}, m_{13}], [m_{21}, m_{22}, m_{23}], [m_{31}, m_{32}, m_{33}]]$, expanding along Row 1:

$$\begin{aligned}\det(M) &= m_{11} \times (m_{22} \times m_{33} - m_{23} \times m_{32}) \\ &\quad - m_{12} \times (m_{21} \times m_{33} - m_{23} \times m_{31}) \\ &\quad + m_{13} \times (m_{21} \times m_{32} - m_{22} \times m_{31})\end{aligned}$$

Each 2×2 block is called a **minor**. The $+/-$ sign is called the **cofactor**. The pattern of signs for Row 1 expansion is always: $+, -, +$

3.3 Sign Pattern for Cofactor Expansion

Position	Sign	Reason
(1,1), (1,3), (2,2), (3,1), (3,3)	+	$(-1)^{(1+1)}, (-1)^{(1+3)}, \text{etc.} = +1$
(1,2), (2,1), (2,3), (3,2)	-	$(-1)^{(1+2)}, (-1)^{(2+1)}, \text{etc.} = -1$

3.4 Worked Eigenvalue Calculation: 2x2 Matrix

Example 3.1 — Find eigenvalues of $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Step 1: Write Given Matrix

$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$. We need to find all values of λ such that $Ax = \lambda x$.

Step 2: Form $A - \lambda I$

Replace each diagonal element by subtracting λ :

$$A - \lambda I = \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix}$$

Why? Because the characteristic equation is $\det(A - \lambda I) = 0$. We first construct $A - \lambda I$.

Step 3: Compute Determinant of $A - \lambda I$

For a 2×2 matrix $\begin{bmatrix} p & q \\ r & s \end{bmatrix}$, $\det = ps - qr$

$$\det(A - \lambda I) = (5-\lambda)(2-\lambda) - (4)(1)$$

Why? This is the standard 2×2 determinant formula: top-left $\times$ bottom-right minus top-right $\times$ bottom-left.

Step 4: Expand $(5-\lambda)(2-\lambda)$ Carefully

$$(5-\lambda)(2-\lambda) = 5 \times 2 - 5 \times \lambda - \lambda \times 2 + \lambda \times \lambda$$

$$= 10 - 5\lambda - 2\lambda + \lambda^2$$

$$= \lambda^2 - 7\lambda + 10$$

Why? We use the FOIL method (First, Outer, Inner, Last) to expand two bracketed terms.

Step 5: Subtract $(4)(1) = 4$

$$\det = \lambda^2 - 7\lambda + 10 - 4 = \lambda^2 - 7\lambda + 6$$

Step 6: Set Equal to Zero (Characteristic Equation)

$$\lambda^2 - 7\lambda + 6 = 0$$

Step 7: Solve the Quadratic Equation

We need two numbers that multiply to give 6 and add to give -7 .

Try: -6 and -1 : $(-6) \times (-1) = 6 \checkmark$ and $(-6) + (-1) = -7 \checkmark$

So: $\lambda^2 - 7\lambda + 6 = (\lambda - 6)(\lambda - 1) = 0$

Therefore: $\lambda = 6$ or $\lambda = 1$

Why? We factored the quadratic. Setting each factor to zero gives the solutions.

Eigenvalues: $\lambda_1 = 1$ and $\lambda_2 = 6$

Concept Summary: The trace of $A = 5+2 = 7 = 1+6$ (sum of eigenvalues $\checkmark$).

Det of $A = 5 \times 2 - 4 \times 1 = 6 = 1 \times 6$ (product of eigenvalues $\checkmark$). These are good checks!

3.5 Algebraic Multiplicity vs Geometric Multiplicity

Algebraic Multiplicity: The number of times a particular eigenvalue appears as a root of the characteristic equation.

Geometric Multiplicity: The number of linearly independent eigenvectors corresponding to that eigenvalue (= dimension of eigenspace).

Important: Geometric multiplicity $\leq$ Algebraic multiplicity always.

SECTION 4: FINDING EIGENVECTORS

4.1 What is an Eigenvector?

Once we have found an eigenvalue λ , we substitute it back into $(A - \lambda I)x = 0$ and solve for the vector x . This vector x is the eigenvector.

4.2 Step-by-Step Process

1. Find all eigenvalues by solving $\det(A - \lambda I) = 0$.
2. For each eigenvalue λ_i , substitute into $(A - \lambda_i I)x = 0$.
3. Write out the system of linear equations.
4. Solve the system (usually by row reduction or back substitution).
5. Express the solution in terms of a free variable (usually t).
6. The resulting non-zero vector is the eigenvector.

Important Note: Eigenvectors are not unique! If x is an eigenvector, then kx (for any non-zero scalar k) is also an eigenvector for the same eigenvalue. We typically choose simple values like $t = 1$ to get a specific eigenvector.

4.3 Worked Example: Finding Eigenvectors

Example 4.1 — Find eigenvectors of $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ (eigenvalues found in Ex 3.1)

Given: $\lambda_1 = 1, \lambda_2 = 6$

For $\lambda_1 = 1$: Compute $A - \lambda_1 I = A - I$

$$A - I = [[5-1, 4], [1, 2-1]] = [[4, 4], [1, 1]]$$

System: $(A-I)x = 0$ gives:

$$4x_1 + 4x_2 = 0 \quad \dots (i)$$

$$1x_1 + 1x_2 = 0 \quad \dots (ii)$$

Why two equations? Because $(A-\lambda I)x = 0$ means matrix times column vector equals zero vector. Each row of the matrix gives one equation.

Solve the system for $\lambda_1 = 1$

From equation (ii): $x_1 + x_2 = 0 \Rightarrow x_1 = -x_2$

Let $x_2 = t$ (free variable, take $t = 1$)

Then $x_1 = -1$

Eigenvector: $x_1 = [-1, 1]$ or equivalently $[1, -1]$

Why a free variable? Because $(A-\lambda I)$ is singular — it has infinite solutions. We parameterize by t and choose convenient values.

For $\lambda_2 = 6$: Compute $A - 6I$

$$A - 6I = [[5-6, 4], [1, 2-6]] = [[-1, 4], [1, -4]]$$

System: $(A-6I)x = 0$ gives:

$$-x_1 + 4x_2 = 0 \quad \dots (i)$$

$$x_1 - 4x_2 = 0 \quad \dots (ii)$$

Solve for $\lambda_2 = 6$

From equation (i): $x_1 = 4x_2$

Let $x_2 = 1$, then $x_1 = 4$

Eigenvector: $x_2 = [4, 1]$

For $\lambda_1=1$: eigenvector = $[-1, 1]$ | For $\lambda_2=6$: eigenvector = $[4, 1]$

4.4 Eigenspace

Eigenspace: The set of all eigenvectors corresponding to an eigenvalue λ , together with the zero vector, forms a subspace called the *eigenspace* of λ . It is the null space of $(A - \lambda I)$.

SECTION 5: PROPERTIES OF EIGENVALUES AND EIGENVECTORS

5.1 List of Key Properties

PROPERTY 1: SUM OF EIGENVALUES = TRACE

The sum of all eigenvalues of a matrix A equals the trace of A (sum of diagonal elements).

$$\lambda_1 + \lambda_2 + \dots + \lambda_n = \text{trace}(A) = a_{11} + a_{22} + \dots + a_{nn}$$

PROPERTY 2: PRODUCT OF EIGENVALUES = DETERMINANT

The product of all eigenvalues equals the determinant of A .

$$\lambda_1 \times \lambda_2 \times \dots \times \lambda_n = \det(A)$$

PROPERTY 3: EIGENVALUES OF A^k

If λ is an eigenvalue of A , then λ^k is an eigenvalue of A^k . The eigenvectors remain the same.

PROPERTY 4: EIGENVALUES OF A^{-1}

If λ is an eigenvalue of A (and $\lambda \neq 0$), then $1/\lambda$ is an eigenvalue of A^{-1} . Same eigenvectors.

PROPERTY 5: EIGENVALUES OF kA

If λ is an eigenvalue of A , then $k\lambda$ is an eigenvalue of kA . Same eigenvectors.

PROPERTY 6: EIGENVALUES OF $A + kI$

If λ is an eigenvalue of A , then $(\lambda + k)$ is an eigenvalue of $A + kI$. Same eigenvectors.

PROPERTY 7: SYMMETRIC MATRIX EIGENVALUES ARE REAL

All eigenvalues of a real symmetric matrix ($A = A^T$) are real numbers.

PROPERTY 8: ORTHOGONAL EIGENVECTORS FOR SYMMETRIC MATRICES

Eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are mutually orthogonal (perpendicular to each other).

PROPERTY 9: ZERO EIGENVALUE AND SINGULARITY

A matrix A has $\lambda = 0$ as an eigenvalue if and only if A is singular ($\det(A) = 0$).

PROPERTY 10: TRIANGULAR MATRIX EIGENVALUES

For a triangular matrix (upper or lower), the eigenvalues are simply the diagonal elements.

5.2 Verification Table

Matrix Property	Eigenvalue Behavior	Eigenvector Behavior
A	λ	x
A^2	λ^2	same x
A^k	λ^k	same x
A^{-1}	$1/\lambda$	same x
kA	$k\lambda$	same x
$A + kI$	$\lambda + k$	same x

A^T	same λ	generally different x
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SECTION 6: DIAGONALIZATION OF A MATRIX

6.1 What is Diagonalization?

Diagonalization: A matrix A is said to be *diagonalizable* if it can be expressed as:

$$A = P \cdot D \cdot P^{-1}$$

where D is a diagonal matrix (all entries are zero except on the main diagonal), and P is an invertible matrix formed by the eigenvectors of A .

Equivalently: $D = P^{-1} \cdot A \cdot P$

6.2 What Goes Into D and P ?

- **D (Diagonal Matrix):** The eigenvalues of A placed on the diagonal. The order must match the columns of P .
- **P (Modal Matrix):** The eigenvectors of A placed as *columns*. Each column corresponds to one eigenvalue.

6.3 When Can We Diagonalize?

CONDITION FOR DIAGONALIZABILITY

An $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors. This happens when:

- All n eigenvalues are distinct (different from each other), OR
- For each repeated eigenvalue, the geometric multiplicity equals the algebraic multiplicity.

6.4 Step-by-Step Diagonalization Process

Example 6.1 — Diagonalize $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Step 1: Find Eigenvalues

From Example 3.1: $\lambda_1 = 1, \lambda_2 = 6$

Step 2: Find Eigenvectors

From Example 4.1: $x_1 = [-1, 1]$ (for $\lambda=1$) and $x_2 = [4, 1]$ (for $\lambda=6$)

Step 3: Form P by placing eigenvectors as columns

$P = \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$ (column 1 = eigenvector for $\lambda=1$, column 2 = eigenvector for $\lambda=6$)

Why as columns? The relationship $A = PDP^{-1}$ requires eigenvectors as columns.
This is the standard convention.

Step 4: Form D with eigenvalues on diagonal

$D = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}$ (eigenvalues in same order as columns of P)

Why same order? Because $D = P^{-1}AP$ and the i -th column of P corresponds to the i -th diagonal element of D .

Step 5: Find P^{-1}

For $P = \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$:

$$\det(P) = (-1)(1) - (4)(1) = -1 - 4 = -5$$

$$P^{-1} = (1/\det(P)) \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= (1/-5) \times \begin{bmatrix} 1 & -4 \\ -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1/5 & 4/5 \\ 1/5 & 1/5 \end{bmatrix}$$

For 2×2 inverse: swap diagonal, negate off-diagonal, divide by determinant.

Step 6: Verify $A = PDP^{-1}$

Compute PD first:

$$PD = \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} -1 \times 1 + 4 \times 0 & -1 \times 0 + 4 \times 6 \\ 1 \times 0 + 1 \times 6 \end{bmatrix} = \begin{bmatrix} -1 & 24 \\ 1 & 6 \end{bmatrix}$$

Then $PD \times P^{-1}$ should give back $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ ✓

$$P = \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}, A = PDP^{-1}$$

6.5 Why is Diagonalization Useful?

- **Computing A^k :** $A^k = PD^kP^{-1}$. Since D is diagonal, $D^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots)$ which is trivial to compute.
- **Solving ODEs:** Decouples systems of differential equations.
- **Understanding matrix structure:** Reveals the fundamental behavior of the transformation.

SECTION 7: CAYLEY-HAMILTON THEOREM

7.1 Statement of the Theorem

CAYLEY-HAMILTON THEOREM

Statement: Every square matrix satisfies its own characteristic equation.

In other words, if $P(\lambda) = 0$ is the characteristic equation of a matrix A , then substituting A for λ in $P(\lambda)$ gives the zero matrix:

$$P(A) = 0$$

Note: The proof of this theorem is beyond the scope of this course. We only need to know the statement and how to apply it.

7.2 What Does This Mean Exactly?

Let's say for a 2×2 matrix, the characteristic equation is:

$$\lambda^2 - 7\lambda + 6 = 0$$

Then by Cayley-Hamilton, we substitute A in place of λ :

$$A^2 - 7A + 6I = 0 \text{ (zero matrix)}$$

This is a *matrix equation* — it says that when you subtract 7 times A and add 6 times the identity matrix from A -squared, you get the zero matrix.

Common Mistake: When substituting A for λ , the constant term c becomes $c \times I$ (the identity matrix), NOT just c . Because you're working with matrices, and a constant must

be a scalar multiple of the identity matrix.

7.3 Verification Example

Example 7.1 — Verify Cayley-Hamilton for $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Step 1: Write the characteristic equation

$$\lambda^2 - 7\lambda + 6 = 0 \text{ (from Example 3.1)}$$

Step 2: By Cayley-Hamilton, substitute A for λ

$$A^2 - 7A + 6I = 0 \text{ (this must be verified)}$$

Step 3: Calculate A^2

$$A^2 = A \times A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$\text{Row 1, Col 1: } 5 \times 5 + 4 \times 1 = 25 + 4 = 29$$

$$\text{Row 1, Col 2: } 5 \times 4 + 4 \times 2 = 20 + 8 = 28$$

$$\text{Row 2, Col 1: } 1 \times 5 + 2 \times 1 = 5 + 2 = 7$$

$$\text{Row 2, Col 2: } 1 \times 4 + 2 \times 2 = 4 + 4 = 8$$

$$A^2 = \begin{bmatrix} 29 & 28 \\ 7 & 8 \end{bmatrix}$$

Matrix multiplication: each element (i,j) = sum of (row i of A) times (column j of A).

Step 4: Calculate $A^2 - 7A + 6I$

$$7A = \begin{bmatrix} 35 & 28 \\ 7 & 14 \end{bmatrix}$$

$$6I = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

$$A^2 - 7A + 6I = \begin{bmatrix} 29 & 28 \\ 7 & 8 \end{bmatrix} - \begin{bmatrix} 35 & 28 \\ 7 & 14 \end{bmatrix} + \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 29-35+6 & 28-28+0 \\ 7-7+0 & 8-14+6 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \text{Zero Matrix } \checkmark$$

Cayley-Hamilton theorem verified: $A^2 - 7A + 6I = 0$

SECTION 8: APPLICATIONS OF CAYLEY-HAMILTON THEOREM

8.1 Why Use Cayley-Hamilton?

The Cayley-Hamilton theorem has two major practical applications:

1. **Finding the inverse of a matrix** without using row operations (faster for exam use)
2. **Finding powers of a matrix** by expressing A^k in terms of lower powers of A

Both applications rely on the same idea: *rearrange* the matrix equation obtained from Cayley-Hamilton to get what you need.

8.2 General Strategy

Key Idea: From the characteristic equation $P(\lambda) = 0$, by Cayley-Hamilton, we get $P(A) = 0$.

Rearranging $P(A) = 0$ allows us to express A^{-1} or A^k in terms of lower powers of A .

SECTION 9: FINDING INVERSE USING CAYLEY-HAMILTON

9.1 Method to Find A^{-1}

Given the characteristic equation:

$$\lambda^n + c_1\lambda^{n-1} + \dots + c_{n-1}\lambda + c_n = 0$$

By Cayley-Hamilton:

$$A^n + c_1A^{n-1} + \dots + c_{n-1}A + c_nI = 0$$

Assuming $c_n \neq 0$ (i.e., $\det(A) \neq 0$, so A is invertible), multiply both sides by A^{-1} :

$$A^{n-1} + c_1A^{n-2} + \dots + c_{n-1}I + c_nA^{-1} = 0$$

Therefore:

$$A^{-1} = -(1/c_n) [A^{n-1} + c_1A^{n-2} + \dots + c_{n-1}I]$$

9.2 Worked Example: 2×2 Inverse

Example 9.1 — Find A^{-1} using Cayley-Hamilton for $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Step 1: Write the characteristic equation

$$\lambda^2 - 7\lambda + 6 = 0$$

$$\text{Here } c_1 = -7, c_2 = 6$$

Step 2: Apply Cayley-Hamilton

$$A^2 - 7A + 6I = 0$$

Step 3: Rearrange to isolate A^{-1}

Multiply both sides on the right by A^{-1} :

$$A - 7I + 6A^{-1} = 0 \text{ (dividing } A^2 \times A^{-1} = A, A \times A^{-1} = I, I \times A^{-1} = A^{-1})$$

$$6A^{-1} = 7I - A$$

$$A^{-1} = (1/6)(7I - A)$$

We divided the equation by 6 (scalar division is straightforward).

Step 4: Compute $7I - A$

$$7I = [[7, 0], [0, 7]]$$

$$7I - A = [[7-5, 0-4], [0-1, 7-2]] = [[2, -4], [-1, 5]]$$

Step 5: Divide by 6

$$A^{-1} = (1/6)[[2, -4], [-1, 5]] = [[2/6, -4/6], [-1/6, 5/6]] = [[1/3, -2/3], [-1/6, 5/6]]$$

Step 6: Verify $A \times A^{-1} = I$

$$[[5, 4], [1, 2]] \times [[1/3, -2/3], [-1/6, 5/6]]$$

$$(1,1): 5 \times (1/3) + 4 \times (-1/6) = 5/3 - 4/6 = 10/6 - 4/6 = 6/6 = 1 \checkmark$$

$$(1,2): 5 \times (-2/3) + 4 \times (5/6) = -10/3 + 20/6 = -20/6 + 20/6 = 0 \checkmark$$

$$(2,1): 1 \times (1/3) + 2 \times (-1/6) = 1/3 - 1/3 = 0 \checkmark$$

$$(2,2): 1 \times (-2/3) + 2 \times (5/6) = -2/3 + 5/3 = 3/3 = 1 \checkmark$$

$$A^{-1} = (1/6)[[2, -4], [-1, 5]]$$

9.3 3×3 Inverse Example

Example 9.2 — Find A^{-1} for $A = [[1, 0, 3], [2, 1, -1], [1, -1, 1]]$ using Cayley-Hamilton

Step 1: Find the characteristic equation

$$\text{Form } A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 3 \\ 2 & 1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)[(1-\lambda)(1-\lambda) - (-1)(-1)] - 0 + 3[2(-1) - (1-\lambda)(1)]$$

$$= (1-\lambda)[(1-\lambda)^2 - 1] + 3[-2 - 1 + \lambda]$$

$$= (1-\lambda)[1 - 2\lambda + \lambda^2 - 1] + 3(\lambda - 3)$$

$$= (1-\lambda)(\lambda^2 - 2\lambda) + 3\lambda - 9$$

$$= \lambda(1-\lambda)(\lambda-2) + 3\lambda - 9$$

$$= \lambda[\lambda - 2 - \lambda^2 + 2\lambda] + 3\lambda - 9$$

$$= \lambda[-\lambda^2 + 3\lambda - 2] + 3\lambda - 9$$

$$= -\lambda^3 + 3\lambda^2 - 2\lambda + 3\lambda - 9$$

$$= -\lambda^3 + 3\lambda^2 + \lambda - 9$$

$$\text{Setting equal to zero: } \lambda^3 - 3\lambda^2 - \lambda + 9 = 0 \text{ (multiply by } -1)$$

Step 2: By Cayley-Hamilton: $A^3 - 3A^2 - A + 9I = 0$

Step 3: Multiply through by A^{-1}

$$A^2 - 3A - I + 9A^{-1} = 0$$

$$9A^{-1} = I + 3A - A^2$$

$$A^{-1} = (1/9)(I + 3A - A^2)$$

Step 4: Compute A^2

$$A^2 = A \times A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\text{Row1,Col1: } 1+0+3=4; \text{ Row1,Col2: } 0+0-3=-3; \text{ Row1,Col3: } 3+0+3=6$$

$$\text{Row2,Col1: } 2+2-1=3; \text{ Row2,Col2: } 0+1+1=2; \text{ Row2,Col3: } 6-1-1=4$$

$$\text{Row3,Col1: } 1-2+1=0; \text{ Row3,Col2: } 0-1-1=-2; \text{ Row3,Col3: } 3+1+1=5$$

$$A^2 = \begin{bmatrix} 4 & -3 & 6 \\ 3 & 2 & 4 \\ 0 & -2 & 5 \end{bmatrix}$$

Step 5: Compute $I + 3A - A^2$

$$3A = \begin{bmatrix} 3 & 0 & 9 \\ 6 & 3 & -3 \\ 3 & -3 & 3 \end{bmatrix}$$

$$I + 3A = \begin{bmatrix} 4 & 0 & 9 \\ 6 & 4 & -3 \\ 3 & -3 & 4 \end{bmatrix}$$

$$\begin{aligned} I + 3A - A^2 &= [[4-4, 0-(-3), 9-6], [6-3, 4-2, -3-4], [3-0, -3-(-2), 4-5]] \\ &= [[0, 3, 3], [3, 2, -7], [3, -1, -1]] \end{aligned}$$

$$A^{-1} = (1/9)[[0, 3, 3], [3, 2, -7], [3, -1, -1]]$$

SECTION 10: POWERS OF MATRIX USING CAYLEY-HAMILTON

10.1 The Core Idea

Using Cayley-Hamilton, we can express any higher power of A in terms of lower powers. This is powerful because computing A^{10} directly would require many matrix multiplications, but using Cayley-Hamilton, we can reduce it dramatically.

10.2 Example: Computing A^4 for a 2×2 Matrix

Example 10.1 — Find A^4 for $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ using Cayley-Hamilton

Step 1: Recall characteristic equation

$$A^2 - 7A + 6I = 0 \text{ (from section 7)}$$

Step 2: Express A^2 in terms of lower powers

$$A^2 = 7A - 6I$$

This means we can always replace A^2 with $(7A - 6I)$.

Why? The Cayley-Hamilton equation gives a direct relationship. From

$$A^2 - 7A + 6I = 0, \text{ we get } A^2 = 7A - 6I.$$

Step 3: Find A^3 by multiplying A^2 by A

$$A^3 = A^2 \times A = (7A - 6I) \times A = 7A^2 - 6A$$

But $A^2 = 7A - 6I$, so substitute:

$$A^3 = 7(7A - 6I) - 6A = 49A - 42I - 6A = 43A - 42I$$

Step 4: Find $A^4 = A^3 \times A$

$$A^4 = (43A - 42I) \times A = 43A^2 - 42A$$

Substitute $A^2 = 7A - 6I$:

$$A^4 = 43(7A - 6I) - 42A = 301A - 258I - 42A = 259A - 258I$$

Step 5: Substitute A to get the numeric matrix

$$259A = [[259 \times 5, 259 \times 4], [259 \times 1, 259 \times 2]] = [[1295, 1036], [259, 518]]$$

$$258I = [[258, 0], [0, 258]]$$

$$A^4 = [[1295 - 258, 1036 - 0], [259 - 0, 518 - 258]] = [[1037, 1036], [259, 260]]$$

$$A^4 = [[1037, 1036], [259, 260]]$$

SECTION 11: INTRODUCTION TO QUADRATIC FORMS

11.1 What is a Quadratic Form?

Definition: A *quadratic form* in n variables $x_1, x_2, \dots, x_n$ is a polynomial where every term has degree exactly 2. That means each term is either x_i^2 or $x_i x_j$ (product of two variables).

Examples of quadratic forms in 2 variables:

- $Q = 3x_1^2 + 5x_2^2$
- $Q = x_1^2 + 4x_1x_2 + 2x_2^2$
- $Q = -x_1^2 + 2x_1x_2 - 3x_2^2$

Examples of quadratic forms in 3 variables:

- $Q = 2x_1^2 + 3x_2^2 + x_3^2 + 4x_1x_2 - 2x_1x_3 + 6x_2x_3$

11.2 Matrix Representation of a Quadratic Form

Any quadratic form can be written as:

$$Q = \mathbf{X}^T \mathbf{A} \mathbf{X}$$

where $\mathbf{X}$ is the column vector of variables and $\mathbf{A}$ is a *symmetric* matrix.

How to Form the Matrix A?

- **Diagonal entries a_{ii} :** The coefficient of x_i^2 directly.
- **Off-diagonal entries a_{ij} ($i \neq j$):** Half the coefficient of $x_i x_j$. That is, $a_{ij} = a_{ji} = (\text{coefficient of } x_i x_j) / 2$.

11.3 Worked Example: Converting to Matrix Form

Example 11.1 — Express $Q = 2x^2 + 5y^2 + 3z^2 + 4xy - 2xz + 6yz$ in matrix form

Step 1: Identify coefficients of each term

Coefficient of $x^2 = 2 \Rightarrow a_{11} = 2$

Coefficient of $y^2 = 5 \Rightarrow a_{22} = 5$

Coefficient of $z^2 = 3 \Rightarrow a_{33} = 3$

Coefficient of $xy = 4 \Rightarrow a_{12} = a_{21} = 4/2 = 2$

Coefficient of $xz = -2 \Rightarrow a_{13} = a_{31} = -2/2 = -1$

Coefficient of $yz = 6 \Rightarrow a_{23} = a_{32} = 6/2 = 3$

Why half? Because $x_i x_j = x_j x_i$. In the symmetric matrix, this cross term is split equally between positions (i,j) and (j,i) , so each gets half.

Step 2: Write the symmetric matrix A

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 2 & 5 & 3 \\ -1 & 3 & 3 \end{bmatrix}$$

Step 3: Write the full quadratic form as $\mathbf{X}^T \mathbf{A} \mathbf{X}$

$$Q = [x, y, z] \times \begin{bmatrix} 2 & 2 & -1 \\ 2 & 5 & 3 \\ -1 & 3 & 3 \end{bmatrix} \times [x, y, z]^T$$

Matrix of QF: $A = \begin{bmatrix} 2 & 2 & -1 \\ 2 & 5 & 3 \\ -1 & 3 & 3 \end{bmatrix}$

11.4 Recovering the Quadratic Form from Matrix

If given A, you can write Q by: $Q = \sum_i a_{ii} x_i^2 + \sum_{i < j} 2a_{ij} x_i x_j$

Remember to multiply off-diagonal entries by 2 when going back to the polynomial form.

SECTION 12: NATURE OF QUADRATIC FORMS

12.1 Why Classify Quadratic Forms?

Different quadratic forms behave very differently. Some are always positive (like a bowl), some are always negative (like an inverted bowl), and some can be either positive or negative depending on the input. This classification is crucial in optimization problems, stability analysis, and physics.

12.2 Definitions

Type	Condition	Eigenvalues	Example
Positive Definite	$Q > 0$ for all $X \neq 0$	All eigenvalues > 0	$x^2 + y^2$
Positive Semi-definite	$Q \geq 0$ for all X	All eigenvalues ≥ 0	x^2
Negative Definite	$Q < 0$ for all $X \neq 0$	All eigenvalues < 0	$-x^2 - y^2$
Negative Semi-definite	$Q \leq 0$ for all X	All eigenvalues ≤ 0	$-x^2$
Indefinite	Q takes both $+$ and $-$ values	Both $+$ and $-$ eigenvalues	$x^2 - y^2$

12.3 Sylvester's Criterion (Leading Principal Minors)

SYLVESTER'S CRITERION FOR POSITIVE DEFINITENESS

A symmetric matrix A is positive definite if and only if all leading principal minors are positive:

$$D_1 = a_{11} > 0$$

$$D_2 = \det([a_{11}, a_{12}], [a_{21}, a_{22}]) > 0$$

$$D_3 = \det(A_{3 \times 3}) > 0$$

... and so on.

12.4 Worked Example: Determining Nature

Example 12.1 — Determine the nature of $Q = 2x^2 + 4y^2 + 6z^2 + 2xy + 2yz$

Step 1: Write the matrix of the quadratic form

$$\text{Coefficient } x^2=2 \Rightarrow a_{11}=2; y^2=4 \Rightarrow a_{22}=4; z^2=6 \Rightarrow a_{33}=6$$

$$xy \text{ has coefficient } 2 \Rightarrow a_{12}=a_{21}=1$$

$$yz \text{ has coefficient } 2 \Rightarrow a_{23}=a_{32}=1$$

$$xz \text{ has coefficient } 0 \Rightarrow a_{13}=a_{31}=0$$

$$A = [[2,1,0],[1,4,1],[0,1,6]]$$

Step 2: Compute leading principal minors

$$D_1 = 2 > 0 \checkmark$$

$$D_2 = \det([[2,1],[1,4]]) = 8-1 = 7 > 0 \checkmark$$

$$D_3 = \det(A) = 2(24-1) - 1(6-0) + 0 = 2 \times 23 - 6 = 46-6 = 40 > 0 \checkmark$$

Leading principal minor D_k is the determinant of the top-left $k \times k$ submatrix.

Step 3: Conclusion

All leading principal minors are positive (> 0), so Q is **Positive Definite**.

Q is Positive Definite (all leading principal minors > 0)

12.5 Using Eigenvalues to Determine Nature

Alternatively, find the eigenvalues of the matrix A :

- All $\lambda_i > 0 \Rightarrow$ Positive Definite
- All $\lambda_i \geq 0$ (at least one $= 0$) $\Rightarrow$ Positive Semi-definite
- All $\lambda_i < 0 \Rightarrow$ Negative Definite
- All $\lambda_i \leq 0$ (at least one $= 0$) $\Rightarrow$ Negative Semi-definite
- Some $\lambda_i > 0$ and some $\lambda_i < 0 \Rightarrow$ Indefinite

SECTION 13: ORTHOGONAL TRANSFORMATION

13.1 What is an Orthogonal Matrix?

Orthogonal Matrix: A square matrix P is called orthogonal if its transpose equals its inverse:

$$P^T = P^{-1} \quad \Rightarrow \quad P^T P = P P^T = I$$

Key properties:

- $\det(P) = \pm 1$
- The columns of P form an orthonormal set (mutually perpendicular unit vectors)
- The rows of P also form an orthonormal set

13.2 Orthonormal Vectors

Orthogonal Vectors: Two vectors u and v are orthogonal if their dot product is zero: $u \cdot v = 0$

Unit Vector: A vector u is a unit vector if its length (norm) equals 1: $\|u\| = \sqrt{u_1^2 + u_2^2 + u_3^2} = 1$

Orthonormal: Vectors that are both orthogonal to each other AND each has unit length.

13.3 Gram-Schmidt Process (Normalizing Eigenvectors)

For a symmetric matrix, eigenvectors corresponding to distinct eigenvalues are automatically orthogonal. We only need to **normalize** them (make each a unit vector) by dividing by their length.

Normalized vector: $u = x / ||x|| = x / \sqrt{x_1^2 + x_2^2 + x_3^2}$

13.4 Orthogonal Diagonalization

SPECTRAL THEOREM

Every real symmetric matrix A can be orthogonally diagonalized:

$$A = P D P^T = P D P^{-1}$$

where P is an orthogonal matrix (columns = normalized eigenvectors) and D is the diagonal matrix of eigenvalues.

13.5 Steps for Orthogonal Diagonalization

1. Find all eigenvalues of A .
2. Find eigenvectors for each eigenvalue.
3. If eigenvalues are distinct, eigenvectors of symmetric matrix are automatically orthogonal.
4. Normalize each eigenvector: divide by its magnitude.
5. Form P with normalized eigenvectors as columns (P is now orthogonal).
6. Form D with eigenvalues on diagonal.
7. Then $D = P^T A P$.

SECTION 14: REDUCTION TO CANONICAL FORM

14.1 What is Canonical Form?

Canonical Form (Normal Form): A quadratic form $Q = X^TAX$ is said to be in canonical form when it contains only squared terms (no cross terms $x_i x_j$). It has the form:

$$Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

where λ_i are the eigenvalues of the matrix A of the quadratic form.

14.2 Process of Reduction

We use the orthogonal transformation $X = PY$, where P is the orthogonal matrix of normalized eigenvectors. This substitution transforms $Q = X^TAX$ to:

$$Q = Y^T(P^TAP)Y = Y^TDY = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

14.3 Complete Worked Example

Example 14.1 — Reduce $Q = 3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to canonical form

Step 1: Write the matrix of the QF

Coefficient $x^2=3, y^2=5, z^2=3$; $xy: -2 \Rightarrow a_{12}=-1$; $yz: -2 \Rightarrow a_{23}=-1$; $zx: +2 \Rightarrow$

$a_{13}=1$

$A = [[3, -1, 1], [-1, 5, -1], [1, -1, 3]]$

Step 2: Find characteristic equation

$$\det(A - \lambda I) = 0$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{bmatrix}$$

Expanding along row 1:

$$= (3-\lambda)[(5-\lambda)(3-\lambda)-1] + 1[(-1)(3-\lambda)-(-1)(1)] + 1[(-1)(-1)-(5-\lambda)(1)]$$

$$= (3-\lambda)[15-8\lambda+\lambda^2-1] + [-3+\lambda+1] + [1-5+\lambda]$$

$$= (3-\lambda)(\lambda^2-8\lambda+14) + (\lambda-2) + (\lambda-4)$$

$$= (3-\lambda)(\lambda^2-8\lambda+14) + 2\lambda-6$$

$$= 3\lambda^2-24\lambda+42 -\lambda^3+8\lambda^2-14\lambda + 2\lambda-6$$

$$= -\lambda^3+11\lambda^2-36\lambda+36$$

$$\text{So: } \lambda^3-11\lambda^2+36\lambda-36 = 0$$

Step 3: Solve characteristic equation

$$\text{Try } \lambda=2: 8-44+72-36=0 \checkmark$$

$$\text{Factor: } (\lambda-2)(\lambda^2-9\lambda+18)=0$$

$$\lambda^2-9\lambda+18=(\lambda-3)(\lambda-6)=0$$

$$\text{Eigenvalues: } \lambda_1=2, \lambda_2=3, \lambda_3=6$$

Step 4: Find eigenvectors for each eigenvalue

$$\text{For } \lambda=2: A-2I = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Row reduce: $R_2 \rightarrow R_2 + R_1$, $R_3 \rightarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

From R_2 : $y=0$; From R_1 : $x=-z+y \dots \Rightarrow x=-z$ (with $y=0$, let $z=1$: $x=-1$)

$$\text{Eigenvector } v_1 = [-1, 0, 1]$$

$$\text{For } \lambda=3: A-3I = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

From row 3: $x=y$; From row 1: $y=z$; So $x=y=z=1$

$$\text{Eigenvector } v_2 = [1, 1, 1]$$

$$\text{For } \lambda=6: A-6I = \begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix}$$

From R_3 : $x-y-3z=0$; From R_2 : $-x-y-z=0 \Rightarrow x=-y-z$

Substituting: $-y-z-y-3z=0 \Rightarrow -2y-4z=0 \Rightarrow y=-2z$; $x=2z-z=z$

Let $z=1$: $x=1$, $y=-2$, $z=1$

Eigenvector $v_3 = [1, -2, 1]$

Step 5: Normalize the eigenvectors

$$\|v_1\| = \sqrt{1+0+1} = \sqrt{2}; p_1 = [-1/\sqrt{2}, 0, 1/\sqrt{2}]$$

$$\|v_2\| = \sqrt{1+1+1} = \sqrt{3}; p_2 = [1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}]$$

$$\|v_3\| = \sqrt{1+4+1} = \sqrt{6}; p_3 = [1/\sqrt{6}, -2/\sqrt{6}, 1/\sqrt{6}]$$

Step 6: Form orthogonal matrix P and write canonical form

$P = [p_1|p_2|p_3]$ (as columns)

Canonical form: $Q = 2y_1^2 + 3y_2^2 + 6y_3^2$

Canonical Form: $Q = 2y_1^2 + 3y_2^2 + 6y_3^2$ (Positive Definite)

SECTION 15: SOLVED EXAMPLES

Set A: Eigenvalue Problems

Example A1 — Find eigenvalues of $A = \begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix}$

Step 1: Given matrix

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix}$$

Step 2: Form $A - \lambda I$

$$\begin{bmatrix} 2-\lambda & 1 \\ 5 & 2-\lambda \end{bmatrix}$$

Step 3: Characteristic equation

$$(2-\lambda)(2-\lambda) - 5 \times 1 = 0$$

$$4 - 4\lambda + \lambda^2 - 5 = 0$$

$$\lambda^2 - 4\lambda - 1 = 0$$

Step 4: Use quadratic formula

$$\lambda = (4 \pm \sqrt{16+4})/2 = (4 \pm \sqrt{20})/2 = 2 \pm \sqrt{5}$$

$$\text{Eigenvalues: } \lambda_1 = 2 + \sqrt{5} \approx 4.24, \lambda_2 = 2 - \sqrt{5} \approx -0.24$$

Example A2 — Find eigenvalues of $A = \begin{bmatrix} 4 & 0 & 0 \\ 3 & 4 & 0 \\ -2 & 6 & 4 \end{bmatrix}$

Step 1: Observe the matrix type

This is a lower triangular matrix (all entries above main diagonal are zero). For triangular matrices, eigenvalues are the diagonal entries!

Step 2: Identify diagonal elements

Main diagonal: 4, 4, 4

Step 3: Verify using characteristic equation

$$\det(A - \lambda I) = (4 - \lambda)(4 - \lambda)(4 - \lambda) = (4 - \lambda)^3 = 0 \Rightarrow \lambda = 4 \text{ (with algebraic multiplicity 3)}$$

Eigenvalue: $\lambda = 4$ (with multiplicity 3)

Example A3 — Find eigenvalues of $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & -1 \end{bmatrix}$

Step 1: Form $A - \lambda I$

$$\begin{bmatrix} 1-\lambda & -1 & 0 \\ 1 & 2-\lambda & 1 \\ 0 & 3 & -1-\lambda \end{bmatrix}$$

Step 2: Expand along first row

$$\begin{aligned} &= (1-\lambda)[(2-\lambda)(-1-\lambda) - 3] - (-1)[1(-1-\lambda) - 0] + 0 \\ &= (1-\lambda)[-2-2\lambda+\lambda+\lambda^2-3] + [-1-\lambda] \\ &= (1-\lambda)[\lambda^2-\lambda-5] + (-1-\lambda) \end{aligned}$$

Step 3: Expand further

$$\begin{aligned} &= \lambda^2 - \lambda - 5 - \lambda^3 + \lambda^2 + 5\lambda - 1 - \lambda \\ &= -\lambda^3 + 2\lambda^2 + 3\lambda - 6 \end{aligned}$$

$$\text{Characteristic equation: } \lambda^3 - 2\lambda^2 - 3\lambda + 6 = 0$$

Step 4: Factor

$$\text{Try } \lambda=2: 8-8-6+6=0 \checkmark$$

$$\text{Factor: } (\lambda-2)(\lambda^2-3)=0$$

$$\lambda=2, \lambda=\sqrt{3}, \lambda=-\sqrt{3}$$

Eigenvalues: $\lambda_1=2, \lambda_2=\sqrt{3}\approx 1.73, \lambda_3=-\sqrt{3}\approx -1.73$

Example A4 — Find eigenvalues of $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$

Step 1: Form $A-\lambda I$ and compute determinant

$$\begin{aligned} \det &= (0-\lambda)[(0-\lambda)(3-\lambda)-(1)(3)] - 1[(0)(3-\lambda)-(1)(1)] + 0 \\ &= -\lambda[-3\lambda+\lambda^2-3] - 1[-1] \\ &= -\lambda^3+3\lambda^2+3\lambda+1 = 0 \end{aligned}$$

Wait, let me expand carefully row 3 = [1, -3, 3], so expanding along column 1 is simpler.

Step 2: Expand $\det(A-\lambda I)$ along row 1

$$\begin{aligned} &= -\lambda[(-\lambda)(3-\lambda)-1\times(-3)] - 1[(0)(3-\lambda)-1\times 1] + 0 \\ &= -\lambda[-3\lambda+\lambda^2+3] - 1[-1] \\ &= -\lambda^3+3\lambda^2-3\lambda+1 = 0 \\ &\Rightarrow \lambda^3-3\lambda^2+3\lambda-1=0 \Rightarrow (\lambda-1)^3=0 \end{aligned}$$

Eigenvalue: $\lambda = 1$ (with algebraic multiplicity 3)

Example A5 — Find eigenvalues of $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Step 1: Characteristic equation

$$\text{trace}(A) = 6+3+3 = 12$$

$$\text{Sum of } 2\times 2 \text{ minors: } (18-1)+(18-4)+(9-4) = 17+14+5 = 36$$

$$\det(A) = 6(9-1)+2(-6+2)+2(2-6) = 48-8-8 = 32$$

$$\text{Characteristic equation: } \lambda^3-12\lambda^2+36\lambda-32=0$$

Step 2: Solve cubic

Try $\lambda=2$: $8-48+72-32=0$ ✓

Factor: $(\lambda-2)(\lambda^2-10\lambda+16)=0$

$(\lambda-2)(\lambda-2)(\lambda-8)=0$

$\lambda=2$ (double), $\lambda=8$

Eigenvalues: $\lambda_1 = 2$ (multiplicity 2), $\lambda_2 = 8$

Set B: Eigenvector Problems

Example B1 — Find eigenvectors of $A = \begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix}$

Eigenvalues from A1: $\lambda_1=2+\sqrt{5}$, $\lambda_2=2-\sqrt{5}$

For $\lambda_1=2+\sqrt{5}$: $A-\lambda_1 I$

$$\begin{bmatrix} 2-(2+\sqrt{5}) & 1 \\ 5 & 2-(2+\sqrt{5}) \end{bmatrix} = \begin{bmatrix} -\sqrt{5} & 1 \\ 5 & -\sqrt{5} \end{bmatrix}$$

Equation 1: $-\sqrt{5}x_1 + x_2 = 0 \Rightarrow x_2 = \sqrt{5}x_1$

Let $x_1=1$: $x_2=\sqrt{5}$

Eigenvector: $[1, \sqrt{5}]$

For $\lambda_2=2-\sqrt{5}$: $A-\lambda_2 I$

$$\begin{bmatrix} \sqrt{5} & 1 \\ 5 & \sqrt{5} \end{bmatrix}$$

Equation 1: $\sqrt{5}x_1 + x_2 = 0 \Rightarrow x_2 = -\sqrt{5}x_1$

Let $x_1=1$: $x_2=-\sqrt{5}$

Eigenvector: $[1, -\sqrt{5}]$

For $\lambda_1=2+\sqrt{5}$: $[1, \sqrt{5}]$ | For $\lambda_2=2-\sqrt{5}$: $[1, -\sqrt{5}]$

Example B2 — Find ALL eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$

Step 1: Characteristic equation

$$A-\lambda I = \begin{bmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{bmatrix}$$

$$\det = (1-\lambda)[(2-\lambda)(2-\lambda)-2] - 2[0-(-1)] + 2[0-(-1)(2-\lambda)]$$

$$= (1-\lambda)[4-4\lambda+\lambda^2-2] - 2 + 2[2-\lambda]$$

$$\begin{aligned}
 &= (1-\lambda)(\lambda^2-4\lambda+2) - 2 + 4-2\lambda \\
 &= \lambda^2-4\lambda+2 - \lambda^3+4\lambda^2-2\lambda + 2-2\lambda \\
 &= -\lambda^3+5\lambda^2-8\lambda+4=0 \\
 &\Rightarrow \lambda^3-5\lambda^2+8\lambda-4=0
 \end{aligned}$$

Step 2: Solve cubic

Try $\lambda=1$: $1-5+8-4=0$ ✓

$$(\lambda-1)(\lambda^2-4\lambda+4)=0$$

$$(\lambda-1)(\lambda-2)^2=0$$

$\lambda=1, \lambda=2$ (double)

Eigenvector for $\lambda=1$: $A-I$

$$[[0,2,2],[0,1,1],[-1,2,1]]$$

R1/2: $[0,1,1]$; R2=R2: $[0,1,1] \rightarrow$ same; R3: $[-1,2,1]$

$$y+z=0 \Rightarrow y=-z; \text{ From R3: } -x+2y+z=0 \Rightarrow x=2y+z=-2z+z=-z$$

Let $z=1$: $x=-1, y=-1, z=1$

Eigenvector: $[-1,-1,1]$

Eigenvector for $\lambda=2$: $A-2I$

$$[[-1,2,2],[0,0,1],[-1,2,0]]$$

From R2: $z=0$; From R3: $-x+2y=0 \Rightarrow x=2y$

Let $y=1$: $x=2, z=0$

Eigenvector: $[2,1,0]$

$$\lambda=1: [-1,-1,1] \mid \lambda=2: [2,1,0]$$

Set C: Diagonalization Problems

Example C1 — Diagonalize $A = [[1, 1, 0], [0, 2, 0], [0, 0, 3]]$

Step 1: Find eigenvalues

Upper triangular matrix! Eigenvalues are diagonal: $\lambda_1=1, \lambda_2=2, \lambda_3=3$

Step 2: Find eigenvectors

For $\lambda=1$: $A-I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$; Equations: $y=0, 2z=0 \Rightarrow y=0, z=0, x=\text{free}$;

Eigenvector: $[1, 0, 0]$

For $\lambda=2$: $A-2I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; $-x+y=0, z=0 \Rightarrow x=y$; Eigenvector:

$[1, 1, 0]$

For $\lambda=3$: $A-3I = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$; $y=0, -2x+y=0 \Rightarrow x=0; z=\text{free}$;

Eigenvector: $[0, 0, 1]$

Step 3: Form P and D

$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example C2 — Diagonalize $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

Step 1: Characteristic equation

$$(3-\lambda)^2 - 1 = 0 \Rightarrow (3-\lambda+1)(3-\lambda-1) = 0 \Rightarrow (4-\lambda)(2-\lambda) = 0$$

$$\lambda_1=2, \lambda_2=4$$

Step 2: Eigenvectors

For $\lambda=2$: $A-2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$; $x+y=0 \Rightarrow y=-x$; Eigenvector: $[1, -1]$

For $\lambda=4$: $A-4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$; $-x+y=0 \Rightarrow y=x$; Eigenvector: $[1, 1]$

Step 3: Diagonalize

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, D = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

Check: $\text{trace}=2+4=6=3+3 \checkmark$; $\text{det}=8=3 \times 3 - 1 \times 1 = 8 \checkmark$

$$P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

Set D: Cayley-Hamilton Problems

Example D1 — Verify Cayley-Hamilton for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Step 1: Characteristic equation

$$\det(A - \lambda I) = (1 - \lambda)(4 - \lambda) - 6 = 4 - 5\lambda + \lambda^2 - 6 = \lambda^2 - 5\lambda - 2 = 0$$

Step 2: Apply Cayley-Hamilton: $A^2 - 5A - 2I = 0$

Step 3: Compute A^2

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$(1,1): 1+6=7; (1,2): 2+8=10; (2,1): 3+12=15; (2,2): 6+16=22$$

$$A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

Step 4: Check $A^2 - 5A - 2I$

$$5A = \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix}; 2I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \checkmark$$

$$\text{Cayley-Hamilton verified: } A^2 - 5A - 2I = 0$$

Example D2 — Find A^{-1} using Cayley-Hamilton for $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$

Step 1: Characteristic polynomial

$$\text{trace}=5, \det(A)=2(2-0)-1(0-0)+1(0-1)=4-1=3$$

$$\text{Sum of } 2 \times 2 \text{ minors: } (2-0)+(4-1)+(2-1)=2+3+1=6$$

$$\text{Char eq: } \lambda^3 - 5\lambda^2 + 6\lambda - 3 = 0 \text{ (verify: adjust signs per expansion)}$$

Step 2: By Cayley-Hamilton

$$A^3 - 5A^2 + 6A - 3I = 0$$

Multiply by A^{-1} :

$$A^2 - 5A + 6I - 3A^{-1} = 0$$

$$A^{-1} = (1/3)(A^2 - 5A + 6I)$$

Step 3: Compute A^2

$$A^2 = [[2, 1, 1], [0, 1, 0], [1, 1, 2]] \times [[2, 1, 1], [0, 1, 0], [1, 1, 2]]$$

$$(1,1)=4+0+1=5; (1,2)=2+1+1=4; (1,3)=2+0+2=4$$

$$(2,1)=0+0+0=0; (2,2)=0+1+0=1; (2,3)=0+0+0=0$$

$$(3,1)=2+0+2=4; (3,2)=1+1+2=4; (3,3)=1+0+4=5$$

$$A^2 = [[5, 4, 4], [0, 1, 0], [4, 4, 5]]$$

Step 4: Compute $A^2 - 5A + 6I$

$$5A = [[10, 5, 5], [0, 5, 0], [5, 5, 10]]; 6I = [[6, 0, 0], [0, 6, 0], [0, 0, 6]]$$

$$[[5, 4, 4], [0, 1, 0], [4, 4, 5]] - [[10, 5, 5], [0, 5, 0], [5, 5, 10]] + [[6, 0, 0], [0, 6, 0], [0, 0, 6]]$$

$$= [[5-10+6, 4-5+0, 4-5+0], [0-0+0, 1-5+6, 0], [4-5+0, 4-5+0, 5-10+6]]$$

$$= [[1, -1, -1], [0, 2, 0], [-1, -1, 1]]$$

$$A^{-1} = (1/3)[[1, -1, -1], [0, 2, 0], [-1, -1, 1]]$$

Example D3 — If $A = [[1, 0], [1, 1]]$, find A^5 using Cayley-Hamilton

Step 1: Characteristic equation

$$(1-\lambda)^2 - 0 = 0 \Rightarrow \lambda^2 - 2\lambda + 1 = 0 \Rightarrow (\lambda-1)^2 = 0 \Rightarrow \lambda=1 \text{ (double)}$$

Step 2: Cayley-Hamilton: $A^2 - 2A + I = 0 \Rightarrow A^2 = 2A - I$

Step 3: Reduce powers step by step

$$A^3 = A \times A^2 = A(2A - I) = 2A^2 - A = 2(2A - I) - A = 3A - 2I$$

$$A^4 = A \times A^3 = A(3A - 2I) = 3A^2 - 2A = 3(2A - I) - 2A = 4A - 3I$$

$$A^5 = A \times A^4 = A(4A - 3I) = 4A^2 - 3A = 4(2A - I) - 3A = 5A - 4I$$

Step 4: Substitute A

$$5A = \begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix}; 4I = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$A^5 = \begin{bmatrix} 5-4 & 0 \\ 5 & 5-4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$$

$$A^5 = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$$

Set E: Complete 3×3 Problems

Example E1 — Find all eigenvalues and eigenvectors of $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Step 1: Characteristic equation

$$\det(A - \lambda I) = \det(\begin{bmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix})$$

Expand along row 3 (has two zeros — easier!):

$$= (1-\lambda) \times \det(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix})$$

$$= (1-\lambda)[(2-\lambda)^2 - 1]$$

$$= (1-\lambda)[4 - 4\lambda + \lambda^2 - 1]$$

$$= (1-\lambda)(\lambda^2 - 4\lambda + 3)$$

$$= (1-\lambda)(\lambda-1)(\lambda-3)$$

$$= -(\lambda-1)^2(\lambda-3) = 0$$

We expanded along row 3 because it has two zeros, making computation much easier — a very useful exam trick!

Step 2: Eigenvalues: $\lambda=1$ (multiplicity 2), $\lambda=3$

Eigenvector for $\lambda=3$: $A-3I$

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

$$\text{From R3: } -2z=0 \Rightarrow z=0$$

$$\text{From R1: } -x+y=0 \Rightarrow x=y$$

$$\text{Let } y=1: x=1, z=0$$

$$\text{Eigenvector: } [1, 1, 0]$$

Eigenvector for $\lambda=1$: $A-I$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Only one independent equation: } x+y+z=0$$

$$\text{Let } y=1, z=0: x=-1 \Rightarrow v_1 = [-1, 1, 0]$$

Let $y=0, z=1$: $x=-1 \Rightarrow v_2 = [-1, 0, 1]$

Geometric multiplicity = 2 (two free variables)

$\lambda=1$: $[-1, 1, 0]$ and $[-1, 0, 1]$; $\lambda=3$: $[1, 1, 0]$

Example E2 — Find all eigenvalues and eigenvectors of $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$

Step 1: Find characteristic equation

$$\text{trace} = 4 + 3 + (-2) = 5$$

Let's compute $\det(A - \lambda I)$ by expanding along row 1:

$$= (4 - \lambda)[(3 - \lambda)(-2 - \lambda) - (2)(-5)] - 6[(1)(-2 - \lambda) - (2)(-1)] + 6[(1)(-5) - (3 - \lambda)(-1)]$$

$$= (4 - \lambda)[-6 - 3\lambda + 2\lambda + \lambda^2 + 10] - 6[-2 - \lambda + 2] + 6[-5 + 3 - \lambda]$$

$$= (4 - \lambda)[\lambda^2 - \lambda + 4] - 6[-\lambda] + 6[-2 - \lambda]$$

$$= (4 - \lambda)(\lambda^2 - \lambda + 4) + 6\lambda - 12 - 6\lambda$$

$$= (4 - \lambda)(\lambda^2 - \lambda + 4) - 12$$

$$= 4\lambda^2 - 4\lambda + 16 - \lambda^3 + \lambda^2 - 4\lambda - 12$$

$$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4$$

$$\text{Setting } = 0: \lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

Step 2: Solve cubic

$$\text{Try } \lambda=1: 1 - 5 + 8 - 4 = 0 \checkmark$$

$$(\lambda - 1)(\lambda^2 - 4\lambda + 4) = 0 \Rightarrow (\lambda - 1)(\lambda - 2)^2 = 0$$

$$\lambda=1, \lambda=2 \text{ (double)}$$

Eigenvector for $\lambda=1$:

$$A - I = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ -1 & -5 & -3 \end{bmatrix}$$

$$R1/3 = [1, 2, 2]; R2 = [1, 2, 2] \text{ (same)}; R3 = [-1, -5, -3]$$

$$\text{From } R1: x + 2y + 2z = 0; \text{ from } R3: -x - 5y - 3z = 0 \Rightarrow x + 5y + 3z = 0$$

Subtract: $3y+z=0 \Rightarrow z=-3y$; $x=-2y-2(-3y)=4y$

Let $y=1$: $x=4, y=1, z=-3$; Eigenvector: $[4, 1, -3]$

Eigenvector for $\lambda=2$:

$$A-2I = \begin{bmatrix} 2 & 6 & 6 \\ 1 & 1 & 2 \\ -1 & -5 & -4 \end{bmatrix}$$

$R_2 = [1, 1, 2]$; From $R_1/2 = [1, 3, 3]$; $3y+2z-(y+2z)=0 \Rightarrow 2y=0 \Rightarrow y=0$; $z=-x/2$ (from R_2 with $y=0$: $x+2z=0$).

Let $x=2$: $z=-1, y=0$; Eigenvector: $[2, 0, -1]$

$$\lambda=1: [4, 1, -3]; \lambda=2: [2, 0, -1]$$

Set F: Quadratic Forms Problems

Example F1 — Determine the nature of $Q = x^2 + 4y^2 + z^2 - 4xy + 2xz - 4yz$

Step 1: Write matrix A

$$a_{11}=1, a_{22}=4, a_{33}=1$$

$$xy: -4 \Rightarrow a_{12}=-2; xz: +2 \Rightarrow a_{13}=1; yz: -4 \Rightarrow a_{23}=-2$$

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

Step 2: Find eigenvalues of A

Observe: rows are proportional ($\text{Row}_2 = -2 \times \text{Row}_1$, $\text{Row}_3 = \text{Row}_1$)

$\Rightarrow \text{rank}(A) = 1 \Rightarrow$ two eigenvalues are zero

$\text{trace} = 6 = \text{sum of eigenvalues}$, so $\lambda_1=0, \lambda_2=0, \lambda_3=6$

$\det(A)=0$ confirms at least one zero eigenvalue.

Step 3: Nature

Eigenvalues: 0, 0, 6 — all ≥ 0 , with some = 0 $\Rightarrow$ **Positive Semi-definite**

Q is Positive Semi-definite (eigenvalues 0, 0, 6)

Example F2 — Find the nature of $Q = -2x^2 - 3y^2 - z^2 + 2xy + 2yz$

Step 1: Matrix of QF

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -3 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

Step 2: Sylvester's criterion

$$D_1 = -2 < 0 \text{ (negative)}$$

$$D_2 = \det(\begin{bmatrix} -2 & 1 \\ 1 & -3 \end{bmatrix}) = 6 - 1 = 5 > 0$$

$$D_3 = \det(A) = -2(3-1) - 1(-1-0) = -4 + 1 = -3 < 0$$

Pattern for 3×3 Negative Definite: $D_1 < 0, D_2 > 0, D_3 < 0 \checkmark$

Q is Negative Definite (Sylvester pattern: -, +, -)

Example F3 — Write out the QF for $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 4 & 0 \\ 2 & 0 & 5 \end{bmatrix}$

Step 1: Extract terms from matrix A

Diagonal: $a_{11}=3, a_{22}=4, a_{33}=5 \Rightarrow$ terms $3x^2, 4y^2, 5z^2$

Off-diagonal (multiply by 2): $a_{12}=1 \Rightarrow 2xy$; $a_{13}=2 \Rightarrow 4xz$; $a_{23}=0 \Rightarrow$ no yz term

$$Q = 3x^2 + 4y^2 + 5z^2 + 2xy + 4xz$$

Example G1 — Reduce $Q = 2x^2 + 2y^2 + 2z^2 + 2xy$ to canonical form by orthogonal transformation

Step 1: Matrix of QF

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Step 2: Characteristic equation

$$\det(A - \lambda I) = \det(\begin{bmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{bmatrix})$$

$$\text{Expand along column 3: } = (2-\lambda) \times \det(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix})$$

$$= (2-\lambda)[(2-\lambda)^2 - 1]$$

$$= (2-\lambda)(3-\lambda)(1-\lambda) = 0$$

$$\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$$

Step 3: Find eigenvectors

$$\text{For } \lambda = 1: A - I = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; z = 0, x + y = 0 \Rightarrow y = -x; v_1 = [1, -1, 0]$$

$$\text{For } \lambda = 2: A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; y = 0, x = 0; z = \text{free}; v_2 = [0, 0, 1]$$

$$\text{For } \lambda = 3: A - 3I = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}; z = 0, -x + y = 0 \Rightarrow y = x; v_3 = [1, 1, 0]$$

Step 4: Verify orthogonality

$$v_1 \cdot v_2 = (1)(0) + (-1)(0) + (0)(1) = 0 \quad \checkmark$$

$$v_1 \cdot v_3 = (1)(1) + (-1)(1) + (0)(0) = 0 \quad \checkmark$$

$$v_2 \cdot v_3 = (0)(1) + (0)(1) + (1)(0) = 0 \quad \checkmark \text{ (all mutually orthogonal)}$$

Step 5: Normalize eigenvectors

$$\|v_1\| = \sqrt{2}; p_1 = [1/\sqrt{2}, -1/\sqrt{2}, 0]$$

$$\|v_2\| = 1; p_2 = [0, 0, 1]$$

$$\|v_3\| = \sqrt{2}; p_3 = [1/\sqrt{2}, 1/\sqrt{2}, 0]$$

Step 6: Write canonical form and orthogonal matrix P

$P=[p_1|p_2|p_3]$ (columns = normalized eigenvectors, in order of eigenvalues)

Canonical Form: $Q = y_1^2 + 2y_2^2 + 3y_3^2$

Canonical Form: $Q = y_1^2 + 2y_2^2 + 3y_3^2$ (Positive Definite)

Example G2 — Reduce $Q = x^2 - y^2$ to canonical form

Step 1: Matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Step 2: A is diagonal, so eigenvalues are diagonal entries: $\lambda_1=1, \lambda_2=-1$

Step 3: Eigenvectors are standard basis: $[1,0]$ and $[0,1]$ (already normalized)

Step 4: Canonical form

$$Q = y_1^2 - y_2^2$$

Canonical Form: $y_1^2 - y_2^2$ (Indefinite: eigenvalues 1 and -1)

Set H: Mixed Advanced Problems

Example H1 — For $A=\begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$, use Cayley-Hamilton to compute $A^6 - 3A^5 + 2A^3$

Step 1: Characteristic equation

$$(3-\lambda)(0-\lambda)-(-2)(1)=0$$

$$-3\lambda+\lambda^2+2=0 \Rightarrow \lambda^2-3\lambda+2=0 \Rightarrow (\lambda-1)(\lambda-2)=0$$

So $\lambda=1,2$; Characteristic eq: $\lambda^2-3\lambda+2=0$

By C-H: $A^2-3A+2I=0 \Rightarrow A^2=3A-2I$

Step 2: Reduce each power

$$A^3=A(A^2)=A(3A-2I)=3A^2-2A=3(3A-2I)-2A=7A-6I$$

$$A^4=A(7A-6I)=7A^2-6A=7(3A-2I)-6A=15A-14I$$

$$A^5=A(15A-14I)=15A^2-14A=15(3A-2I)-14A=31A-30I$$

$$A^6=A(31A-30I)=31A^2-30A=31(3A-2I)-30A=63A-62I$$

Step 3: Compute $A^6-3A^5+2A^3$

$$=(63A-62I)-3(31A-30I)+2(7A-6I)$$

$$=63A-62I-93A+90I+14A-12I$$

$$=(63-93+14)A+(-62+90-12)I$$

$$=-16A+16I$$

$$=16(I-A)=16[[1-3, 0+2], [0-1, 1-0]]=16[[-2,2], [-1,1]]=[[-32,32], [-16,16]]$$

$$A^6-3A^5+2A^3 = [[-32,32], [-16,16]]$$

Example H2 — Show that $A = [[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$ is orthogonal

Step 1: Compute A^T

$$A^T = [[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$$

Step 2: Compute $A^T \times A$

$$(1,1): \cos^2\theta + \sin^2\theta = 1$$

$$(1,2): -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$$

$$(2,1): \sin\theta\cos\theta - \cos\theta\sin\theta = 0$$

$$(2,2): \sin^2\theta + \cos^2\theta = 1$$

$$A^T A = [[1,0], [0,1]] = I \checkmark$$

Step 3: $\det(A)=\cos^2\theta+\sin^2\theta=1$ (positive orthogonal = rotation matrix)

A is orthogonal since $A^T A = I$ and $\det(A) = 1$

Example H3 — Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and state the nature of the associated quadratic form

Step 1: Characteristic equation

$$\det(A - \lambda I) = \det(\begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix})$$

Expand row 1:

$$= -\lambda(\lambda^2 - 1) - 1(-\lambda - 1) + 1(1 + \lambda)$$

$$= -\lambda^3 + \lambda + \lambda + 1 + 1 + \lambda$$

$$= -\lambda^3 + 3\lambda + 2$$

$$\text{Setting } = 0: \lambda^3 - 3\lambda - 2 = 0$$

Step 2: Solve

$$\text{Try } \lambda = -1: -1 + 3 - 2 = 0 \quad \checkmark$$

$$(\lambda + 1)(\lambda^2 - \lambda - 2) = 0$$

$$(\lambda + 1)(\lambda - 2)(\lambda + 1) = 0$$

$$(\lambda + 1)^2(\lambda - 2) = 0$$

$$\lambda = -1 \text{ (double)}, \lambda = 2$$

Eigenvector for $\lambda = 2$: $A - 2I = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

$R1 + R2 + R3 = [0, 0, 0]$ means dependent; From $R1$ and $R2$: $-2x + y + z = 0$, $x - 2y + z = 0$

Subtract: $-3x + 3y = 0 \Rightarrow x = y$; From $R1$: $-2x + x + z = 0 \Rightarrow z = x$

Eigenvector: $[1, 1, 1]$

Eigenvector for $\lambda = -1$: $A + I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Only one equation: $x+y+z=0$

Two free variables: Let $y=1, z=0 \Rightarrow x=-1: v_1 = [-1, 1, 0]$

Let $y=0, z=1 \Rightarrow x=-1: v_2 = [-1, 0, 1]$

Nature of QF:

Eigenvalues $-1, -1, 2$ — both positive and negative $\Rightarrow$ **Indefinite**

$\lambda=-1: [-1, 1, 0], [-1, 0, 1] \mid \lambda=2: [1, 1, 1] \mid$ QF Nature: Indefinite

Example H4 — Verify Cayley-Hamilton and use it to find A^{-1} for $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & -1 \\ 4 & -3 & 1 \end{bmatrix}$

Step 1: Characteristic equation of A

trace $= 2+1+1=4$

Sum of 2×2 minors: $(2+1)+(2-4)+(2-0) = 3-2+2=3$... let me expand $\det(A-\lambda I)$ fully.

$\det = (2-\lambda)[(1-\lambda)(1-\lambda)-3] + 1[(0)(1-\lambda)+4] + 1[0-4(1-\lambda)]$... expanding carefully:

$$= (2-\lambda)(\lambda^2-2\lambda-2) + (4) + (-4+4\lambda)$$

$$= (2-\lambda)(\lambda^2-2\lambda-2) + 4\lambda$$

Let me use the formula: $\lambda^3 - \text{tr}(A)\lambda^2 + (\text{sum minors})\lambda - \det(A) = 0$

$$\text{tr}(A)=4; \det(A)=2(1-3)+1(0+4)+1(0-4)=-4+4-4=-4$$

Sum of 2×2 minors: $M_{11}=1-3=-2; M_{22}=2-4=-2; M_{33}=2-0=2; \text{Total}=-2$

Char eq: $\lambda^3 - 4\lambda^2 - 2\lambda + 4 = 0$ (need to verify)

Try $\lambda=2$: $8-16-4+4=-8 \neq 0$... try $\lambda=-1$: $-1-4+2+4=1 \neq 0$

Using expanded form: $\lambda^3 - 4\lambda^2 + 1\lambda + 2$... let me recompute.

$$\det(A)=2(1-3)+1(0+4)+1(0-4)=-4+4-4=-4; \text{tr}=4$$

Char eq: $\lambda^3 - 4\lambda^2 + (\text{sum})\lambda - (-4) = 0$

$$\lambda^3 - 4\lambda^2 + (\text{sum})\lambda + 4 = 0; \text{ with sum of principal } 2 \times 2 \text{ minors} = -2$$

Final: $\lambda^3 - 4\lambda^2 - 2\lambda + 4 = 0 \Rightarrow$ wait, $-\det = +4$, so: $\lambda^3 - 4\lambda^2 - 2\lambda + 4 = 0$

Step 2: By Cayley-Hamilton: $A^3 - 4A^2 - 2A + 4I = 0$

Multiply by A^{-1} :

$$A^2 - 4A - 2I + 4A^{-1} = 0$$

$$A^{-1} = (1/4)(4A + 2I - A^2)$$

Step 3: Compute A^2

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & -1 \\ 4 & -3 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & -1 \\ 4 & -3 & 1 \end{bmatrix}$$

$$(1,1) = 4 + 0 + 4 = 8; (1,2) = -2 - 1 - 3 = -6; (1,3) = 2 + 1 + 1 = 4$$

$$(2,1) = 0 + 0 - 4 = -4; (2,2) = 0 + 1 + 3 = 4; (2,3) = 0 - 1 - 1 = -2$$

$$(3,1) = 8 - 0 + 4 = 12; (3,2) = -4 - 3 - 3 = -10; (3,3) = 4 + 3 + 1 = 8 \dots$$

$$A^2 = \begin{bmatrix} 8 & -6 & 4 \\ -4 & 4 & -2 \\ 12 & -10 & 8 \end{bmatrix} \text{ (approximate; verify independently)}$$

Step 4: Compute $4A + 2I - A^2$

$$4A = \begin{bmatrix} 8 & -4 & 4 \\ 0 & 4 & -4 \\ 16 & -12 & 4 \end{bmatrix}$$

$$4A + 2I = \begin{bmatrix} 10 & -4 & 4 \\ 0 & 6 & -4 \\ 16 & -12 & 6 \end{bmatrix}$$

$$4A + 2I - A^2 = \begin{bmatrix} 10 - 8 & -4 + 6 & 4 - 4 \\ 0 + 4 & 6 - 4 & -4 + 2 \\ 16 - 12 & -12 + 10 & 6 - 8 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 0 \\ 4 & 2 & -2 \\ 4 & -2 & -2 \end{bmatrix}$$

$$A^{-1} = (1/4) \begin{bmatrix} 2 & 2 & 0 \\ 4 & 2 & -2 \\ 4 & -2 & -2 \end{bmatrix}$$

Example H5 — If the eigenvalues of A are 2, 3, -1 , find the eigenvalues of (a) A^2 (b) $2A + I$ (c) A^{-1} (d) A^T

Using Properties of Eigenvalues (without computing)

(a) Eigenvalues of A^2

By Property 3: λ^2 gives eigenvalues 4, 9, 1

(b) Eigenvalues of $2A + I$

By Properties 5 and 6: $2\lambda+1$ gives eigenvalues 5, 7, -1

(c) Eigenvalues of A^{-1}

By Property 4: $1/\lambda$ gives eigenvalues $1/2$, $1/3$, -1

(d) Eigenvalues of A^T

By Property (transpose): same eigenvalues 2, 3, -1

(a) 4,9,1 | (b) 5,7, -1 | (c) $1/2,1/3,-1$ | (d) 2,3, -1

SECTION 16: EXERCISE QUESTIONS

Solve all problems step-by-step. Show every calculation. Do not skip steps.

Part A: 2-Mark Questions (Short Answer)

Q1. Define eigenvalue and eigenvector with one example each.

Q2. State the characteristic equation of a 2×2 matrix A and what it represents.

Q3. If $\lambda_1=3$ and $\lambda_2=-2$ are eigenvalues of A , what is $\det(A)$ and $\text{trace}(A)$?

Q4. Write the characteristic equation of $A = \begin{bmatrix} 7 & 0 \\ 0 & -3 \end{bmatrix}$.

Q5. What are the eigenvalues of the identity matrix $I_{3 \times 3}$?

Q6. If λ is an eigenvalue of A , what is the eigenvalue of A^3 ?

Q7. State the Cayley-Hamilton theorem.

Q8. What is the condition for a matrix to be orthogonal?

Q9. Define a quadratic form in 2 variables. Give one example.

Q10. When is a quadratic form called positive definite?

Q11. If A is a 3×3 symmetric matrix with eigenvalues 1, 4, 9, what is the canonical form of $Q = X^T A X$?

Q12. Find the eigenvalues of $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (diagonal matrix).

Q13. What is the matrix representation of $Q = 4x^2 + 9y^2$?

Q14. If all eigenvalues of a matrix A are negative, what is the nature of $Q = X^T A X$?

Q15. State two properties of eigenvalues of a symmetric matrix.

Part B: 5-Mark Questions (Full Problems)

Q16. Find all eigenvalues and eigenvectors of $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Verify sum and product of eigenvalues.

Q17. Find all eigenvalues and eigenvectors of $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$.

Q18. Verify Cayley-Hamilton theorem for $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$.

Q19. Using Cayley-Hamilton, find A^{-1} for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Q20. Find A^4 using Cayley-Hamilton for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Q21. Diagonalize $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

Q22. Find all eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$.

Q23. Write the quadratic form $Q = 3x^2 + 2y^2 + 5z^2 + 4xy - 6yz + 2xz$ in matrix form and determine its nature.

Q24. Reduce $Q = x^2 + 5y^2 + z^2 + 2xy + 6yz + 2xz$ to canonical form by orthogonal transformation.

Q25. Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$.

Q26. Using Cayley-Hamilton, find A^{-1} and verify for $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$...
[Note: if singular, state why inverse doesn't exist.]

Q27. Reduce $Q = 10x^2 + 2y^2 + 5z^2 - 4xy - 10xz + 6yz$ to canonical form.

Q28. Find eigenvalues of $A = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 5 & 4 \\ -4 & 4 & 3 \end{bmatrix}$. Is A positive definite?

Q29. Diagonalize the symmetric matrix $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$.

Q30. Verify Cayley-Hamilton for $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$ and find A^3 .

Q31. If $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, find A^{10} using eigenvalue method.

Q32. Show that $A = \begin{bmatrix} (1/\sqrt{2}), 1/\sqrt{2} \\ -1/\sqrt{2}, 1/\sqrt{2} \end{bmatrix}$ is orthogonal.

Q33. Find the orthogonal transformation that diagonalizes $A = \begin{bmatrix} 5, -4 \\ -4, 5 \end{bmatrix}$.

Q34. Determine the nature of $Q = 2x^2 + 4y^2 + 3z^2$ using Sylvester's criterion (for the matrix $A = \begin{bmatrix} 2, 0, 0 \\ 0, 4, 0 \\ 0, 0, 3 \end{bmatrix}$).

Q35. Find eigenvalues and state the nature of $Q = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx$.

Part C: Additional Challenge Problems

Q36. Find A^6 using Cayley-Hamilton for $A = \begin{bmatrix} 1, 1 \\ 0, 1 \end{bmatrix}$.

Q37. Find the eigenvalues and eigenvectors of $A + 2I$ if eigenvalues of $A = \begin{bmatrix} 1, 2 \\ 0, 3 \end{bmatrix}$ are known.

Q38. Determine whether $A = \begin{bmatrix} 1, 2, 3 \\ 2, 5, 6 \\ 3, 6, 10 \end{bmatrix}$ is positive definite.

Q39. Reduce $Q = 4x_1^2 + 3x_2^2 + 8x_3^2 + 12x_1x_2 - 4x_1x_3$ to canonical form and state its nature.

Q40. Prove that the eigenvalues of a real symmetric matrix are always real by using a suitable 2×2 example.

Q41. Find A^{-1} using C-H theorem for $A = \begin{bmatrix} 0, 1, 2 \\ 1, 2, 3 \\ 3, 1, 1 \end{bmatrix}$.

Q42. Determine the nature of $Q = x^2 + y^2 - 2xy$.

Q43. Diagonalize $A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$.

Q44. For matrix A with trace = 6 and det = 8, and two eigenvalues $\lambda_1=2$, $\lambda_2=2$, find the third eigenvalue.

Q45. Reduce $Q = 8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 + 4x_1x_3 + 8x_2x_3$ to canonical form.

Q46. Find the eigenvalues of A^2+3A-I if A has eigenvalues 0, 1, -2 .

Q47. If A is an $n \times n$ matrix, show that the product of eigenvalues equals $\det(A)$ for a 3×3 numeric example.

Q48. Verify: if $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, the geometric multiplicity of eigenvalue $\lambda=2$ is 1, not 2.

Q49. Given $Q = x^2+4y^2+9z^2$, write the matrix, find eigenvalues, and state the canonical form (which is Q itself in this case—explain why).

Q50. Find A^8 for $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ using Cayley-Hamilton theorem.

SECTION 17: THEOREMS, FORMULA SHEET & QUICK REVISION

17.1 Complete Theorem Summary

THEOREM 1: CAYLEY-HAMILTON THEOREM

Every square matrix satisfies its own characteristic equation. If $P(\lambda)=0$ is the characteristic equation of A , then $P(A)=0$ (the zero matrix).

THEOREM 2: SPECTRAL THEOREM

Every real symmetric matrix can be orthogonally diagonalized. That is, there exists an orthogonal matrix P such that $P^TAP = D$ (diagonal matrix of eigenvalues).

THEOREM 3: TRACE-EIGENVALUE RELATION

For any $n \times n$ matrix A : sum of eigenvalues = trace(A) = sum of diagonal entries.

THEOREM 4: DETERMINANT-EIGENVALUE RELATION

For any $n \times n$ matrix A : product of eigenvalues = det(A).

THEOREM 5: SYLVESTER'S CRITERION

A symmetric matrix A is positive definite if and only if all leading principal minors $D_1, D_2, \dots, D_n$ are positive.

THEOREM 6: ORTHOGONAL TRANSFORMATION PRESERVES FORM

If $X = PY$ is an orthogonal transformation (P orthogonal), then $Q = X^TAX$ transforms to $Q = Y^TDY$ where $D = P^TAP$ is the diagonal matrix of eigenvalues.

THEOREM 7: REAL EIGENVALUES OF SYMMETRIC MATRICES

All eigenvalues of a real symmetric matrix are real numbers. Moreover, eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.

17.2 Complete Formula Sheet

EIGENVALUE FORMULAS

Formula	Description
$\det(A - \lambda I) = 0$	Characteristic Equation
$\lambda^2 - (\text{tr } A)\lambda + \det(A) = 0$	Characteristic eq. for 2×2
$\lambda^3 - (\text{tr } A)\lambda^2 + (\text{sum of } 2\times 2 \text{ minors})\lambda - \det(A) = 0$	Characteristic eq. for 3×3
$\Sigma \lambda_i = \text{tr}(A)$	Sum of eigenvalues
$\Pi \lambda_i = \det(A)$	Product of eigenvalues
$(A - \lambda I)x = 0$	Eigenvector equation
Eigenvalues of $A^k = \lambda^k$	Power of matrix
Eigenvalues of $A^{-1} = 1/\lambda$	Inverse of matrix
Eigenvalues of $kA = k\lambda$	Scalar multiple
Eigenvalues of $A+kI = \lambda+k$	Shift of matrix

DIAGONALIZATION FORMULAS

Formula	Description
$A = PDP^{-1}$	Diagonalization formula
$D = P^{-1}AP$	Diagonal form
$A^k = PD^kP^{-1}$	Powers via diagonalization
$P^{-1} = (1/\det P) \times \text{adj}(P)$	Inverse of P
$D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$	Diagonal matrix
$P = [x_1 \mid x_2 \mid \dots \mid x_n]$	Modal matrix (eigenvectors as columns)

CAYLEY-HAMILTON FORMULAS

Operation	Formula
Theorem	$P(A) = 0$ where $P(\lambda)$ is characteristic poly
2×2 Inverse	From $A^2 - (\text{tr})A + (\det)I = 0 \Rightarrow A^{-1} = (1/\det)[\text{tr} \cdot I - A]$
3×3 Inverse	Multiply $P(A)=0$ by A^{-1} , isolate A^{-1}
Power Reduction	Use $A^2 = -c_1A - c_2I$ (for 2×2) to reduce all higher powers

QUADRATIC FORMS FORMULAS

Concept	Formula / Rule
QF representation	$Q = X^TAX$, A symmetric
Diagonal entry	$a_{ii} = \text{coeff of } x_i^2$
Off-diagonal entry	$a_{ij} = (1/2)(\text{coeff of } x_i x_j)$

Canonical form	$Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$
Orthogonal transformation	$X = PY, P^T A P = D$
Normalizing eigenvector	$p_i = x_i / \ x_i\ $
Norm of vector [a,b,c]	$\text{sqrt}(a^2 + b^2 + c^2)$

NATURE OF QUADRATIC FORMS

Nature	Eigenvalue Condition	Sylvester (3×3)
Positive Definite	All $\lambda_i > 0$	$D_1 > 0, D_2 > 0, D_3 > 0$
Pos. Semi-Definite	All $\lambda_i \geq 0$	$D_k \geq 0$ for all k
Negative Definite	All $\lambda_i < 0$	$D_1 < 0, D_2 > 0, D_3 < 0$
Neg. Semi-Definite	All $\lambda_i \leq 0$	$D_k \leq 0$ alternating
Indefinite	Mixed + and -	No fixed pattern

17.3 Quick Revision Notes

EXAM TRICKS & MEMORY AIDS

- "STD" for characteristic equation:** Sum (trace), pTproduct (det), and Degree (n)
- Triangular/diagonal matrix:** Eigenvalues = diagonal entries. No calculation needed!
- Zero eigenvalue:** If $\det(A) = 0$, then $\lambda = 0$ is one eigenvalue
- Repeated eigenvalue:** Write $(\lambda - r)^k$ where r is value and k is multiplicity
- C-H inverse shortcut (2×2):** $A^{-1} = (1/\det(A))[\text{tr}(A) \cdot I - A]$
- Checking eigenvector:** Multiply Ax — if result = λx , it's correct
- Orthogonal check:** Compute $P^T P$ — must equal I

8. **Normalize before orthogonal matrix:** Divide each eigenvector by its norm

COMMON MISTAKES TO AVOID:

1. Forgetting to write λI when forming $A - \lambda I$ (not just subtracting λ from diagonal)
2. In C-H: writing constant term as c instead of cI (it must be scalar times identity matrix)
3. In QF matrix: forgetting to take HALF of cross-term coefficients
4. Using non-normalized eigenvectors in orthogonal matrix P (columns must have unit length)
5. Confusing algebraic and geometric multiplicity (they need not be equal)
6. Expanding the wrong row/column in determinant (always choose row with most zeros)
7. Forgetting sign alternation (+, -, +) in cofactor expansion
8. Not checking if $\det(A) \neq 0$ before finding A^{-1}

17.4 Step-by-Step Checklist for Exams

PROBLEM-SOLVING CHECKLIST

Problem Type	Steps	Verify
Find Eigenvalues	1. Form $A - \lambda I$ 2. Compute $\det$ 3. Solve polynomial	Sum=trace, Product= $\det$
Find Eigenvectors	1. Substitute λ into $(A - \lambda I)x = 0$ 2. Solve system 3. Express in terms of free variable	Verify $Ax = \lambda x$
Diagonalize	1. Find eigenvalues 2. Find eigenvectors 3. Form P and D	Verify $A = PDP^{-1}$
C-H Inverse	1. Write char eq 2. Apply C-H 3. Multiply by A^{-1} 4. Isolate A^{-1}	Verify $AA^{-1} = I$
C-H Powers	1. Write char eq 2. Express A^2 in lower terms 3. Reduce iteratively	Check with direct multiplication

QF $\rightarrow$ Matrix	1.Diagonal: coeff of x_i^2 2.Off-diag: half coeff of $x_i x_j$	Verify $X^T A X$ gives original QF
Nature of QF	1.Find eigenvalues of A 2.Check signs	Or use Sylvester's criterion
Canonical Form	1.Find λ s and eigenvectors 2.Normalize 3.Write $Q = \sum \lambda_i y_i^2$	Nature follows from signs of eigenvalues

Final Words: Mastery of this unit comes from practice. Follow every step described in this document. Never skip steps — especially in exams. The examiner rewards *method*, not just the final answer. Understanding WHY each step is done will make you confident in any variation of the problem.

Key formula to memorize: $\det(A - \lambda I) = 0$. Everything else in this unit flows from this single equation.